11 Poisson Distributions

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location.

Example 11.1. Recall the matching problem from Example 6.3, Exercise 7.3, Exercise 8.2, and Exercise 9.2. There are n distinct objects that are shuffled and placed uniformly at random in n distinct spots with one object per spot, and X is the number of correct matches of object with spot. Note: n is a fixed and known number, but we are considering different values of it.

We have seen that $\mathrm{E}(X) = 1$ for any n and that unless n is small (e.g., $n \leq 6$), $\mathrm{Var}(X) = 1$ and the approximate distribution of X is given by Table 11.1.

Starting with $x = 0$, describe the distribution of X in terms of relative likelihoods:

1. 1 is [blank] times as likely as 0

2. 2 is [blank] times as likely as 1

3. 3 is [blank] times as likely as 2

4. 4 is [blank] times as likely as 3

Table 11.1: Poisson(1) distribution, the approximate distribution of X, the number of matches in the matching problem for general n and uniformly random placement of objects in spots.

x	P(X=x)
0	0.36788
1	0.36788
2	0.18394
3	0.06131
4	0.01533
5	0.00307
6	0.00051
7	0.00007

5. 5 is [blank] times as likely as 4

6. 6 is [blank] times as likely as 5

7. In general, x is [blank] times as likely as $x-1$, for $x = 1, 2, 3, ...$

- A random variable X has a **Poisson distribution** with parameter $\mu > 0$ if the possible values are $0, 1, 2, ...$ and the pmf follows the

$$\text{Poisson}(\mu) \text{ pattern:} \quad x \text{ is } \frac{\mu}{x} \text{ as likely as } x-1 \qquad (\text{for } x = 1, 2, 3, ...)$$

- As a function of $x = 0, 1, 2, ...$, a Poisson(μ) pmf increases for $x < \mu$ and then decreases for $x > \mu$
  - If μ is a whole number, then the Poisson pmf assigns the same probability to the values μ and $\mu - 1$.
  - If $\mu < 1$ then $x = 0$ has the highest probability and the pmf decreases from there.

- If X has a Poisson(μ) distribution then

$$\begin{aligned} \mathrm{E}(X) &= \mu \\ \mathrm{Var}(X) &= \mu \end{aligned}$$

Example 11.2. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Technically, there is no fixed upper bound on what X can be, so mathematically it is convenient to consider $0, 1, 2, \ldots$ as the possible values of X. Assume that X has a Poisson distribution with parameter 2.4.

1. Interpret 2.4 in context.

2. Identify what the Poisson(2.4) patterns implies in context.

3. Sketch a plot of the distribution of X.

4. Assuming that the probability that X is greater than 9 is negligible, make a table representing the distribution of X.

5. The pmf of X is

$$\mathrm{P}(X = x) = e^{-2.4}\frac{2.4^x}{x!}, \quad x = 0, 1, 2, \ldots$$

   Use the pmf to compute $\mathrm{P}(X = 0)$.

6. Use the pmf to compute $\mathrm{P}(X = 1)$.

7. Use the pmf to compute $\mathrm{P}(X = 2)$.

8. Use the pmf to construct a table and plot representing the distribution of X. Does it match your previous work?

9. Use the pmf to compute $\mathrm{E}(X)$ and verify that it is 2.4.

10. Compute and interpret $\mathrm{P}(X \leq 2)$.

- A discrete random variable X has a Poisson distribution with parameter $\mu > 0$ if and only if its probability mass function p_X satisfies

$$p_X(x) = \frac{e^{-\mu}\mu^x}{x!}, \quad x = 0, 1, 2, \ldots$$

- The Poisson pattern implies that the shape of a Poisson pmf as a function of x is given by $\mu^x/x!$.

$$p(x) = p(0)\frac{\mu^x}{x!}, \qquad x = 0, 1, 2, \ldots$$

- The constant $p(0) = e^{-\mu}$ simply renormalizes the pmf so that the probabilities sum[1] to 1.

Example 11.3. Let X be the number of home runs hit (in total by both teams) in a randomly selected Major League Baseball game. Assume that X has a Poisson(2.4) distribution.

1. In what ways is this like the Binomial situation?

2. In what ways is this NOT like the Binomial situation?

3. In what sense are home runs "relatively rare"?

- Binomial models have several restrictive assumptions that might not be satisfied in practice
- Poisson models are more flexible and are often used to model the distribution of random variables that count the number of "relatively rare" events that occur over a certain interval of time/in a certain location
- **Poisson paradigm.** Let $A_1, A_2, \ldots, A_n$ be a collection of n events. Suppose event i occurs with marginal probability $p_i = \mathrm{P}(A_i)$. Let $N = \mathrm{I}_{A_i} + \mathrm{I}_{A_2} + \cdots + \mathrm{I}_{A_n}$ be the random variable which counts the number of the events in the collection which occur. Suppose
  - n is "large" (but not necessarily fixed),

[1] Recall the series expansion $e^{\mu} = \sum_{x=0}^{\infty} \frac{\mu^x}{x!}$

  - $p_1, \dots, p_n$ are "comparably small" (but not necessarily the same), and
  - the events $A_1, \dots, A_n$ are "not too dependent" (but not necessarily independent).
- Then N has an approximate Poisson distribution with parameter $\mathrm{E}(N) = \sum_{i=1}^n p_i$.

11.1 Exercises

Exercise 11.1. Continuing Example 11.1. Explain why X in the matching problem has a Poisson distribution when n is large.

Exercise 11.2. Suppose that the number of fatal crashes in SLO County in a week follows, approximately, a Poisson(0.53) distribution, independently from week to week. Let X be the number of fatal crashes in a period of 4 consecutive weeks (close enough to a "month").

1. Make an educated guess for $\mathrm{E}(X)$.

2. Compute $\mathrm{P}(X = 0)$.

3. Compute $\mathrm{P}(X = 1)$.

4. How could you use simulation to approximate the distribution of X?

5. Use simulation to approximate the distribution of X. Does the approximate distribution follow (approximately) the Poisson pattern?

6. Use a Poisson pmf to compute $\mathrm{P}(X = 5)$.

- **Poisson aggregation.** If X and Y are independent, X has a Poisson(μ_X) distribution, and Y has a Poisson(μ_Y) distribution, then $X + Y$ has a Poisson($\mu_X + \mu_Y$) distribution.
- If component counts are independent and each has a Poisson distribution, then the total count also has a Poisson distribution.